Experiment 8.2

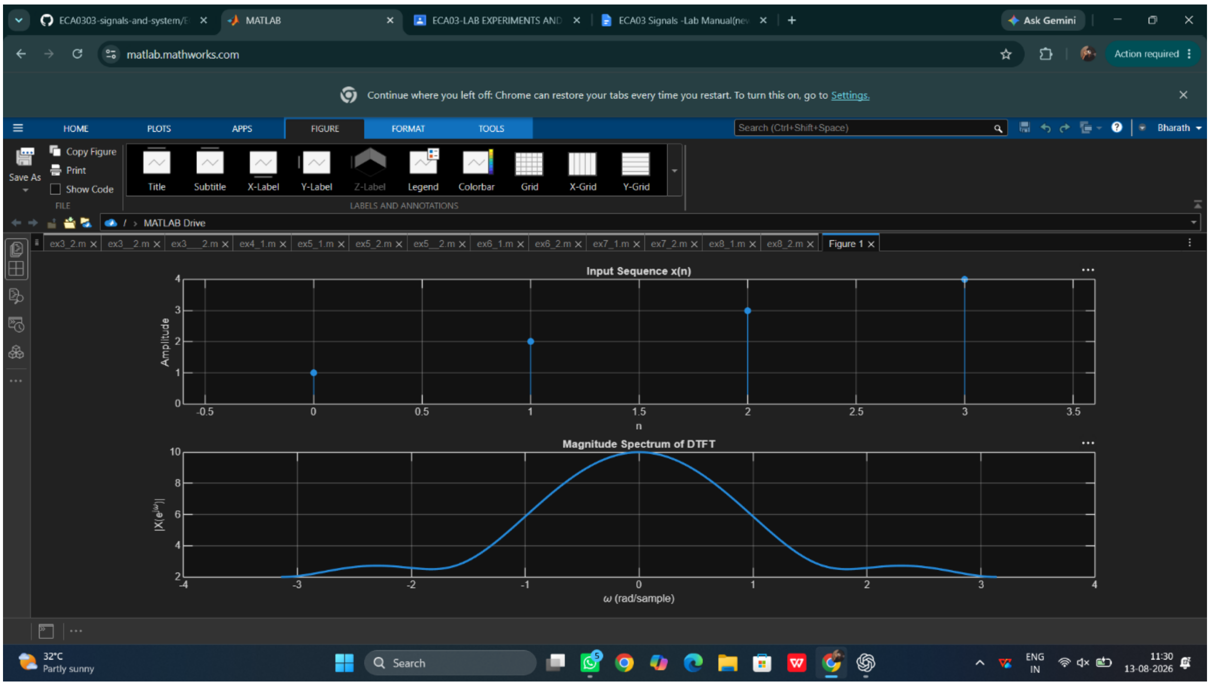
Discrete Time Fourier Transform

Program:

```
% Discrete-Time Fourier Transform (DTFT)

clc;

clear;

close all;

% Input sequence

x = [1 2 3 4];

% Frequency range

w = -pi:0.01:pi;

% DTFT computation

X = zeros(size(w));

for k = 1:length(w)

for n = 1:length(x)

X(k) = X(k) + x(n)*exp(-1j*w(k)*(n-1));

end

end

% Plotting

figure;

subplot(2,1,1);

stem(0:length(x)-1,x,'filled');

grid on;

title('Input Sequence x(n)');

xlabel('n');

ylabel('Amplitude');
```

```
subplot(2,1,2);

plot(w,abs(X),'LineWidth',2);

grid on;

title('Magnitude Spectrum of DTFT');

xlabel('\omega (rad/sample)');

ylabel('|X(e^{j\omega})|');
```

Output Waveform:



Result:

The Discrete-Time Fourier Transform (DTFT) of the sequence x(n)={1,2,3,4} was computed using MATLAB. The magnitude spectrum was obtained and plotted, illustrating the frequency domain representation of the discrete-time signal.